

SHUNWEI (DAVID) DU

(718) 844-0811 // du.david@nyu.edu // [linkedin.com/in/shunwei-david-du/](https://www.linkedin.com/in/shunwei-david-du/)

EDUCATION

- Expected 12/25 **NEW YORK UNIVERSITY** New York, NY
The Courant Institute of Mathematical Sciences
M.S. in Mathematics in Finance
- **Expected Coursework:** stochastic calculus, time series analysis, scientific computing, risk and portfolio management, dynamic asset pricing, algorithmic trading, equity derivatives
- 09/20 - 05/24 **NEW YORK UNIVERSITY** New York, NY
B.A. With Honors in Computer Science and Mathematics
- **Coursework:** linear algebra, probability & statistics, ordinary differential equations, real analysis, numerical analysis, data structures, algorithms, machine learning, data management and analysis
 - **Honors:** Dean's List (4 years), Cum Laude

EXPERIENCE

- 06/24 - 08/24 **QILIN INVESTMENT** Shanghai, China
(Hedge fund with \$4B AUM)
Quantitative Research Intern (Python)
- Generated alpha signals with alternative datasets, including sell-side analyst estimates and sentiment analysis on market news and public opinion; achieved information ratio of 1.5+
 - Built daily-balanced timing signals using MSCI Crowding Models for Barra risk factors, supported by technical indicators, resulting in Sharpe ratio increase of 8.37% for strategy
 - Analyzed Level 2 data (e.g., order book and trade data from institutional investors) to generate signals for long/short equity selection within strategies; improved annual return by 13.68%
- 06/23 - 08/23 **LONGQI INVESTMENT** Hangzhou, China
(Asset management firm with \$2B AUM)
Quantitative Research Intern (Python)
- Developed index rebalance trading strategy based on semiannual rebalancing of CSI 300 and CSI 500 Indexes, incorporating market cap and trading volume criteria; achieved Sharpe ratio of 1.96
 - Designed algorithmic trading strategy using natural language processing on alternative datasets from due diligence reports of listed companies; achieved 16.31% annualized excess return
 - Created 10+ alpha signals by processing 1-minute intraday data and statistical techniques like regression to model market behavior, each achieving information ratio of 1.6+

PROJECTS

- 08/23 - Present **NEW YORK UNIVERSITY** New York, NY
CS Honors Thesis: Evaluating Vision and Language Models for Radiology (Python)
- Established comprehensive evaluation baseline incorporating BERT-similarity scores, prompt-generated GPT evaluations, and Chexpert labeling
 - Conducted comparative analysis of Med-Flamingo, GPT-4/4o, and Gemini, providing insights into performance of commercial and open-source models
- 01/23 - 05/23 **Fit-finder Application Development (Python, HTML, CSS)**
- Designed and developed Fit-finder web application that provided outfit recommendations based on restaurant dress codes
 - Used PyTorch, Torchvision, and FashionCLIP for classification of garment data from Farfetch; developed search engine for users to specify and receive outfit recommendations
- 10/22 - 12/22 **Robot Hand Fingertip Positions Prediction with RGBD Images (Python)**
- Trained convolutional neural network model to identify robot hands' fingertip movements with RGB+Depth images; achieved mean-squared loss of <0.003 in fingertip predictions

COMPUTATIONAL SKILLS / OTHER

Programming Languages: Python, Java, C/C++, SQL, MATLAB, HTML, CSS, Shell Scripting

Languages: English (fluent), Mandarin (native)

Activities: President of NYU Zen Buddhism Club, Web Design Grader at NYU